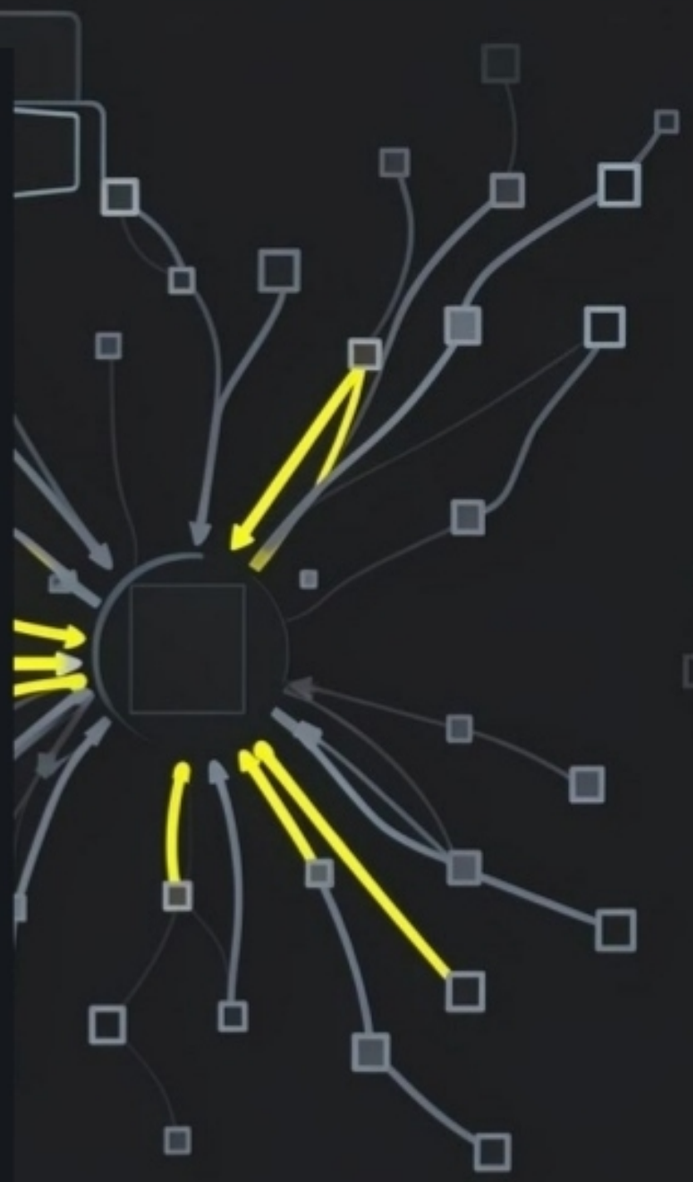
GLUCOTWIN: SYSTEM ARCHITECTURE & DATA PIPELINE

End-to-end technical overview of real-time metabolic simulation and predictive AI modeling.

- Objective: Establish a scalable, low-latency framework for continuous metabolic monitoring and digital twin forecasting.
- Core Architecture Pillars: Modular client interfaces, secure API ingestion gateways, digital twin inference engines, and clinical data persistence.
- Design Standard: High availability, secure data encryption, and responsive web performance.



HIGH-LEVEL SYSTEM ARCHITECTURE FLOW

01

Client Tier (Frontend)

React and Tailwind CSS web interface (hosted via Bolt) handling user interaction, lifestyle parameter inputs, and real-time visualization graphs.

02

API & Ingestion Layer

Secure backend routing endpoints managing continuous metric synchronization, user sessions, and payload validation.

03

Digital Twin Inference Engine

Mathematical and machine learning simulation modules processing carbohydrate/fat/protein inputs against baseline profiles to generate glucose projections.

04

Clinical Storage & Output Layer

Encrypted database managing historical logs, time-series metrics, JSON state files, and clinical PDF report generation.

END-TO-END DATA PROCESSING PIPELINE

01

Step 1 - Ingestion

User modifies simulation controls (carbs, protein, fat, post-meal walk duration, sleep) or logs health markers on the web dashboard.

02

Step 2 - Validation & Processing

The API gateway sanitizes inputs and routes parameters to the digital twin metabolic simulation core.

03

Step 3 - AI Simulation

The engine computes multi-hour glycemic curves, contrasting baseline vs. simulated scenarios within a safe 70–140 mg/dL target window.

04

Step 4 - Real-Time Feedback

Instantaneous rendering of CGM clinical metrics (Time-in-Range, SD, CV%, eA1c) and automated AI risk advisory outputs.

TECHNOLOGY STACK & DATA SECURITY

Frontend & UI	Modern web stack (React, HTML5, responsive CSS framework, hosted on Bolt environment).
Backend & Simulation Logic	Python / FastAPI-powered processing core handling time-series physiological simulation algorithms.
Data Security & Compliance	HTTPS encrypted transport layers, privacy-first session architecture, and secure local/cloud state management.